ADOBE TECHNICAL REPORT

Toward Physics-Faithful Video Generation

A depth-guided generator, a human-annotated flaw benchmark, and an auditable multi-tool critic

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REPORT STATUS AND SCOPE

Scope. This report describes three components built toward physics-faithful video generation: (1) a generator that renders a physics simulation into photorealistic video through a depth/structure control channel; (2) a human-annotated benchmark of prompt-versus-video flaws; and (3) an auditable multi-tool critic that detects such flaws. The three are designed to compose into a loop in which the critic's judgment refines the generator; **that closed loop is future work and is not evaluated here** (section 5).

What is and is not reported. We report the design of each component and the measured *descriptive* statistics of the artifacts (clip counts, flaw counts, generation configuration, model identifiers). We do *not* report critic accuracy, flaw coverage, false-alarm rates, a comparison between critic variants, or a learned video-quality score for the generator; those are performance claims deferred to a dedicated evaluation. Generator validation in this report is qualitative (visual inspection and adherence to the control signal), not a quantitative quality metric. Section 6 states the measurement protocol these results will follow.

Status vocabulary. BUILT present and exercised in the codebase; PARTIAL present with limited semantics; STUB present but inactive; NOT INTEGRATED built but not wired into the full loop; PROPOSED designed, not implemented.

Abstract

Text-to-video generators produce clips of high visual quality that nonetheless violate physical common sense—objects interpenetrate, motion ignores gravity, quantities fail to conserve—and often simply fail to depict what the prompt requested. Making generated video *physically faithful* requires three things at once: a generator that can be steered by physical structure rather than appearance alone; a way to measure, reliably and auditably, when a clip is wrong; and ground-truth data about the flaws real generators make. This report presents one component for each.

First, a **physics-guided generator**: a rigid-body simulation is rendered into a per-frame depth (or silhouette) control video and passed to the native control branch of a Wan 2.2-VACE video-diffusion backbone, so that scene geometry and motion are driven by the simulation while the text prompt supplies appearance. Second, a **human-annotated flaw benchmark**: 1,500 AI-generated clips carry human misalignment annotations—each with a severity, a time span, per-frame bounding boxes, the violated prompt fragment, and a free-text rationale—from which we derive a frozen, physics-focused evaluation set. Third, an **auditable multi-tool critic**: a served vision-language reasoner decomposes a prompt into checkable claims, routes each to a frozen open perception specialist that returns localized evidence only, and combines

the outcomes with deterministic rules into a three-valued verdict (*plausible, implausible, abstain*). The evaluation target is human labels, and no closed or proprietary model produces the critic's verdict.

The intended end state is a loop in which the critic supplies the reward that refines the generator. We describe each component and its honest implementation status, and we specify the human-label evaluation protocol under which the critic and the loop will be measured; comparative performance numbers are deferred to that evaluation.

1 Introduction

1.1 Physical faithfulness as the goal

State-of-the-art text-to-video models render strikingly realistic clips that still break physical common sense and often omit or distort what the prompt asked for. Physics-IQ quantifies the first problem directly: strong visual realism coexists with a large gap in physical understanding on its evaluated models and tasks [38]. Closing that gap is our long-term objective, and it decomposes into three coupled needs.

- **A steerable generator.** A model that follows *physical structure*—the geometry and motion of a scene—rather than reproducing the surface statistics of its training data.
- **A reliable, auditable detector of flaws.** A measurement instrument that says when a clip is wrong *and* shows why, at the level of individual evidence—because that judgment is the reward signal any improvement loop consumes.
- **Ground truth about real flaws.** A benchmark of the mistakes real generators make, annotated by people, against which any detector is measured.

1.2 Three contributions and the loop they form

This report contributes one component for each need and states how they are intended to compose.

1. **A physics-guided video generator** (section 2) that converts a rigid-body simulation into a depth or silhouette control video and drives a Wan 2.2-VACE diffusion backbone's native control branch, separating physical structure (from the simulation) from appearance (from the prompt).
2. **A human-annotated flaw benchmark** (section 3): a 1,500-clip corpus of AI-generated video with localized human misalignment labels, and a frozen physics-focused subset used as the evaluation target.
3. **An auditable multi-tool critic** (section 4) that decomposes a prompt into claims, routes each to an open perception specialist returning evidence only, and combines the evidence deterministically into a three-valued verdict.

The end state (section 5) is a closed loop: the benchmark trains and evaluates the critic, and the critic's verdict becomes the reward that refines the generator. *That loop is future work*; this report builds and characterizes its three parts.

Figure placeholder — generation pipeline.

rigid-body simulation → depth / silhouette control video → Wan-VACE control branch (control = depth, mask = all-white, prompt = appearance) → photorealistic clip.

Structure from the simulation; appearance from the prompt. To be rendered.

Figure 1: Physics-guided generation. The control video carries scene geometry and motion; the prompt carries appearance. **[TODO: replace with real frames: simulation, depth control, generated clip, side by side]**

1.3 Non-claims

We do not claim the closed generator–critic loop exists or that the critic has yet improved any generated video. We do not report critic accuracy, flaw coverage, false-alarm rate, or a comparison between critic variants; those await the evaluation of section 6. We do not report a learned video-quality score for the generator—its validation here is qualitative. And we claim novelty for none of the individual building blocks (structure-conditioned generation, prompt decomposition, specialist routing, three-valued abstention); the contribution is their integration toward physical faithfulness.

2 Physics-Guided Video Generation

2.1 Overview

The generator answers a specific question: can a video-diffusion model be made to follow the *geometry and motion* of a known physical scene, while a text prompt supplies only appearance? Our approach renders a physics simulation into a control video and feeds it to a diffusion backbone that accepts an explicit control channel, so structure comes from the simulation and texture from the prompt (fig. 1). The pipeline is: a rigid-body simulation → a per-frame depth (or silhouette) render → generation through the backbone's native control branch.

2.2 Backbone and control mechanism

The backbone is Wan 2.2-VACE-Fun-14B, a two-expert mixture-of-experts video-diffusion model with a native control branch, run through the `WanVACEPipeline` interface; because the two expert stacks exceed the memory of a single 80 GB accelerator, the experts are CPU-offloaded during generation. Generation is conditioned by passing the pre-computed control video into the pipeline's control input and an all-white mask into its mask input—the instruction to "generate the entire frame under the guidance of the control"—together with the text prompt for appearance and a single scalar *conditioning scale* governing how strongly the control branch binds. Crucially, the backbone consumes a *pre-computed* control video and does not re-estimate depth from an RGB source; the geometry is exactly the simulation's. This is a control-branch mechanism, not image-to-image denoising: it is neither an SDEdit-style noised re-synthesis [35] of a rendered source nor a copy

Setting	Value
Backbone	Wan 2.2-VACE-Fun-14B (two-expert MoE), diffusers `WanVACEPipeline`, bf16
Memory	experts CPU-offloaded (two-expert weights exceed an 80 GB accelerator)
Control signal	MuJoCo z-buffer depth (globally normalized); object silhouette for flat scenes
Conditioning	control video + all-white mask + text prompt; no reference image
Resolution	848×480 (flow-shift 3.0)
Frames / rate	81 frames at 16 fps
Sampler / steps	UniPC, 50 steps
Guidance scale	5.0
Conditioning scale	1.0 (swept over $\{0.5, 0.8, 1.0\}$)

Table 1: Deployed generation configuration, from the generation scripts. A small checkpoint provides a pre-flight sanity gate and is not used for final clips.

of source RGB, which is what distinguishes it from conditioning schemes that leak the source's synthetic appearance into the result.

2.3 Rendering the control signal

The control video is produced from a single simulation pass. We simulate rigid-body dynamics in MuJoCo [47] and, in the same pass, read the renderer's z-buffer for depth and its segmentation buffer for a silhouette. Depth is mapped to a near-bright grayscale under the inverse-depth convention the control branch expects, and—importantly—normalized *globally* across the whole clip from robust percentiles, not per frame, so that a receding object dims consistently instead of being re-stretched every frame. Most scenes use depth; flat tabletop scenes, where a single depth plane carries little signal, use the object silhouette instead. Structure-conditioned generation of this kind—driving synthesis from an explicit geometric control rather than from text alone—follows the lineage of ControlNet for images [57] and motion/structure control for video [25, 49]; here the control is a physically simulated depth field.

2.4 Pipeline and configuration

Table 1 lists the operating configuration. Generation runs as a staged GPU job: control videos are rendered, a small checkpoint provides a fast sanity gate, the 14B checkpoint generates the clips (data-parallel when more than one GPU is available), and the clips are streamed to storage.

2.5 What it achieves, and how we validated it

Generation was validated *qualitatively*: by visual inspection of the output clips and by a lightweight color-blob trajectory tracker that checks whether the generated object follows the control's path. There is no learned or quantitative video-quality score (no FVD, no human-preference model, no critic score) applied to this line, and we make no such claim. Under this qualitative bar, single-object scenes—a bouncing ball, a projectile arc—render cleanly with the object following the simulated trajectory. Multi-object interaction (e.g. a billiards break) is *not* solved: neither the deployed backbone nor the evaluated alternates produce a clean multi-object collision, and we report it as an open problem rather than a result.

2.6 Design choices

Two choices define the deployed recipe and are stated as design decisions. (1) **Native control over source re-synthesis.** A control branch that consumes a pre-computed depth video keeps the generator from importing a rendered source's synthetic look. (2) **No reference image.** The pipeline supports an optional appearance reference, but the deployed recipe omits it; including it degraded single-object trajectories in our inspection. A separate, in-progress study probes *when* during denoising the depth control can be released while the motion it has already induced is retained; its measurements are high-variance and not yet conclusive, and we do not report it as a settled result.

3 A Human-Annotated Video-Flaw Benchmark

3.1 The corpus

The benchmark is built from a corpus of **1,500 AI-generated video clips**, each carrying human annotations of where the video fails to match its prompt. The clips are AI-generated; the generating model is not recorded in the data, and we do not attribute them to a specific system. Every clip has at least one human-flagged flaw (table 2). This corpus is distinct from existing physical-commonsense benchmarks such as VideoPhy and its action-centric successor [5, 6], which we treat as separate transfer targets rather than as the annotation source here.

3.2 Human annotation schema

Each human flaw is a localized, typed record. The annotator marks, per flaw: a **severity** and **confidence**; a **time span** (start/end frame); **per-frame bounding boxes** localizing the flaw in space; the **prompt fragment** the video violates; and a free-text **rationale**. This is a richer object than a clip-level preference score: it says *what* is wrong, *where* in the frame, *when* in time, and *against which part of the prompt*. These fields are the human ground truth. A separate, coarse *scope* and *category* tag on each flaw is assigned automatically by a language-model ensemble, not by the human annotator, and we label it as machine-derived wherever it is used—it is convenience metadata, not part of the human ground truth. We record annotations as single-annotator; the corpus does not carry multiple independent annotations per flaw, and we make no inter-annotator-agreement claim.

3.3 The physics-focused evaluation set

The critic is evaluated not on all 1,500 clips but on a frozen, physics-focused subset, selected so that the flaws exercise physical and compositional reasoning rather than, say, subtitle typos. A physics-relevance filter keeps **606 clips with 1,107 flaws** as the full evaluation set, from which a

Set	Clips	Human flaws
Full corpus (all AI-generated, all annotated)	1,500	2,582
Physics-relevant pool	606	1,171
Full evaluation set (frozen)	606	1,107
Routine evaluation core (frozen)	150	306
continuity stratum within the core	29	85
Disjoint training pool (excludes the frozen sets)	711	—

Table 2: Benchmark statistics, from the dataset manifests. The full evaluation set drops a small number of audio-scoped flaws relative to the physics-relevant pool. The training pool excludes every clip in the frozen evaluation sets.

smaller **frozen core of 150 clips and 306 flaws** is drawn for routine measurement. The core is fixed by a recorded seed and date so that every experiment is scored against exactly the same target; a continuity-focused stratum (29 clips, 85 flaws) is tracked within it. Table 2 summarizes the counts. We are explicit that the evaluation set is the 606-clip / 1,107-flaw derivative (with a frozen 150-clip / 306-flaw core), not the full 1,500-clip corpus; conflating the two would overstate the evaluation. We are also explicit that the 150-clip / 306-flaw core is used as an *optimization set*: system and prompt decisions are repeatedly made against it, so coverage on it measures fit to that core rather than generalization, and a generalization claim requires a separately sequestered held-out set that has not influenced system design, which is future work.

3.4 Recall-oriented evaluation and a held-out training pool

Because every clip in the pool carries at least one flaw and there are no clean clips, the benchmark measures *recall*—did the critic find the human flaws—and cannot on its own measure precision or false-alarm rate; a "flag-everything" detector would score perfectly on recall. We state this limitation plainly and address false alarms in the protocol of section 6. For any learning that consumes this data, a training pool of 711 clips is held strictly disjoint from the frozen evaluation sets, so the evaluation core is never trained on.

4 The Auditable Multi-Tool Critic

4.1 Overview

The critic takes a prompt q and a video v and is designed to return a verdict in $\{\text{plausible}, \text{implausible}, \text{abstain}\}$ with a trace: the claims it checked, the evidence each tool produced, and the rule that combined them. A single holistic "is this plausible?" score would hide which observation drove the verdict and could confound photorealism with prompt satisfaction. Instead the critic decomposes: a served vision-language *reasoner* turns the prompt into small checkable claims and grounds each to the measurements it needs; frozen *perception specialists* return localized evidence only; and a *deterministic core* interprets most of that evidence with typed three-valued predicates and rolls the per-claim outcomes into the clip verdict. The learned parts are the reasoner and the specialists; the aggregation is deterministic and inspectable.

Capability	Open backend	Evidence returned
Object presence	LLMDet open-vocabulary detector [13]	presence; bounding box
Object count	CountGD-Box counter [2, 3] (SAM 2.1 internal)	cardinality of a named set
Kinematics	point tracking, TAPNext-style [59] + kinematic predicates	per-frame positions; static/-moving, direction, speed
Action timing	TimeLens-family temporal grounder [58]	time window of a named event
Spatial relation	unified SAM 3 masks [8] + monocular depth + reasoning	depth-ordered / image-plane relations
Holistic	*no tool by design* [20]	—
Audio†	FlexSED sound-event detection [18]	sound events with timestamps

Table 3: Perception specialists and the open backend wired to each. In the routed configuration the object, count, kinematics, action, and spatial-relation backends execute as stages; the object and count adapters declare stronger primaries that are not on the executed path, so the running backend is shown. The relation capability is served by a unified mask-plus-depth reasoning arm, and no holistic scoring tool is used in the verdict path by design. †The audio backend is wired in the typed-plan path and is not yet exercised in the routed configuration.

4.2 The reasoner

One served vision-language model—Qwen3-VL-30B-A3B-Instruct [39], BF16 weights (no quantization), tensor-parallel over two GPUs—performs the decomposition and grounding and runs three reasoner-mediated hard sub-checks (section 4.6). It is the only learned component that reasons over the whole prompt.

4.3 Perception specialists

Each claim is routed to a frozen open specialist that measures one thing and returns only that measurement (table 3). Two honesty notes are built into the table. First, some adapters *declare* a stronger primary backend that is not the one actually executed; we list the backend that runs. Second, there is no holistic scoring tool in the verdict path by design: a learned holistic video scorer is kept out so that the verdict is built from localized evidence and deterministic rules, though such a scorer could still serve as an external baseline.

4.4 Evidence-only tool contract

Specialists are frozen and constrained by a schema that forbids verdict-, label-, and mismatch-shaped fields: a specialist reports *what it measured* (a box, a count, a track, a time window, a depth), not *whether the prompt is satisfied*. A fail-closed adapter rejects any output that violates the schema, so a malformed or verdict-shaped output cannot enter the accepted evidence record, and each accepted measurement carries the prompt span and entity/event identifier that requested it. This guarantees that no specialist *directly emits* the verdict; it does not make a learned measurement semantically neutral, and a confidently wrong measurement can still mislead a downstream predicate.

4.5 Three-valued semantics and deterministic roll-up

Every claim resolves to supported, contradicted, or unknown, and unknown is first-class: when evidence is insufficient or ambiguous the critic abstains rather than guess. The gates are asymmetric—a positive, resolvable signal is required to assert contradicted, whereas a failed or empty detection becomes unknown, never proof of absence. The clip roll-up is a separate deterministic step: if any claim is contradicted, the clip is *implausible*; otherwise, if all claims are supported, it is *plausible*; otherwise the critic abstains. "Plausible" therefore means only that no checked claim was contradicted—a summary of the executed plan, not a certificate of complete physical verification.

4.6 Reasoner-mediated hard sub-checks

Three checks resist full symbolization and remain reasoner-mediated on tool-localized inputs: object subtype identity (is the detected object the specific kind named), dense action-event recall (did a named event occur, densely over time), and depth-ordered spatial relations. Each runs a fresh reasoner pass over evidence a tool has already localized, under the same asymmetric gate. We name them explicitly rather than imply a pipeline that is more symbolic than it is.

4.7 Compiling claims into an executable measurement plan

Routing could be a flat claim-to-tool table. To make the orchestration inspectable, we instead compile each prompt into a *verification plan*: a small, checkable program that names the measurements the claims require. The plan is validated before any tool runs, equivalent measurements are executed once, and the resulting evidence is combined by fixed rules. This subsection describes how a plan is written, checked, routed, and shared. It concerns *plan construction only*: nothing here tests video verdicts, flaw coverage, or false accusations, and nothing here bears on whether plan-based execution outperforms independent per-claim verification (section 6).

4.7.1 A restricted language for measurement plans

Each claim is represented by **one line of code** in a small, typed language—the code-as-plan idea used in visual programming systems [17, 46], here narrowed to a verification vocabulary. For claim c_N the planner writes `check(cN, <expr>)`, where the expression is a single call or several joined by `and`. The reasoner does not see the video at this stage; it is writing down *what would have to be measured*, not answering. Representing the plan this way avoids cross-referenced identifiers spread across several generated tables and allows the plan to be checked mechanically before execution—planner-format fragility being a reported difficulty in tool-controller systems [43]. Table 4 lists the callable operations and the check each becomes.

4.7.2 Validating a plan before it runs

Because the plan is code, it can be checked mechanically before any tool is invoked. Each line is parsed; a line that does not parse, or that is not of the form `check(cN, …)`, is rejected rather than guessed at. Only calls from the vocabulary in table 4 are permitted, and each is checked for arity

Operation	Meaning	Becomes
`judge("assertion")`	a dense-frame pass over the clip decides the assertion	event occurrence
`exists("object")`	the object is visible at some point	existence
`eq(count("phrase"), N)`	exactly N are visible	count
`before(window(A), window(B))`	A ends before B	temporal order
`during(window(A), window(B))`	A falls within B	temporal containment
`rel("A||B", relation)`	spatial relation between two segmented objects	spatial
`traj("phrase", relation, target)`	tracked motion of an object	trajectory
`sound("description")`	the described sound is asserted	occurrence†

Table 4: The planner's measurement vocabulary. `count` and `window` are structural and legal only inside `eq`, `before`, or `during`. Relations are drawn from fixed vocabularies (for example *behind*, *in front of*, *above*, *contained in* for space; *moves toward*, *rises then falls*, *accelerates* for motion), and the three directed motion relations require an explicit target. †The plan-based path has no dedicated sound-event check: `sound` is routed to the general verifier, so on this path it does not constitute an audio measurement. The audio specialists of table 3 are used in the routed configuration (section 4.7.7).

and argument shape—a count must wrap a `count` term, a spatial relation must name two objects and a relation from the fixed vocabulary, a directed motion relation must carry a target. Logical negation is accepted only directly around the three assertion-style operations, because a negated measurement would require uncertainty handling the downstream rules do not provide.

Failures are handled at two granularities, and neither is silent. If an entire line is unusable, the claim falls back to direct verification by the general verifier, and that fallback is *charged*—counted in the plan's cost like any other call, so that fallback never looks free. If a single term is unusable—an invented relation, a directed motion without a target—only that term degrades to a generalist judgment; the substitution is recorded and explicitly counted as *not* a measurement, so that a plan cannot inflate its measurement rate by mislabeling salvaged terms. Every plan carries this accounting with it: how many calls, which claims fell back, which terms were salvaged.

4.7.3 Several checks per claim, and negation

A claim rarely reduces to a single test. Because an expression may join several calls, each conjunct becomes its own sub-claim with its own measurement, and the claim's outcome is composed from theirs by a fixed rule: **if any part is contradicted the claim is contradicted; if every part is supported the claim is supported; otherwise the claim is left unresolved**. Negation is applied at this composition step rather than pushed into the measurement layer, so the measurement rules never have to reason about negated queries.

Composition is what lets one sentence be checked in more than one way. A claim that a ball triggers a cascade of a thousand mousetraps yields both a judgment that the cascade occurs *and* a count check against the number. Under the composition rule, a measured count below the stated number is by itself sufficient to contradict the claim even when the occurrence part is supported—a claim carrying only an occurrence check cannot express that.

4.7.4 Routing by what the claim says

Which measurement a claim receives is decided by its content rather than by a blanket rule. A numeral introduces a count check alongside the occurrence judgment; spatial prepositions introduce a spatial relation; directed or shaped motion introduces a trajectory check; ordering words introduce a temporal relation; a described sound introduces the audio operation. Anything else—and anything the planner is unsure about—is sent to the general verifier. That fallback guarantees the claim remains executable; it does not guarantee a correct verdict. Relations outside the fixed vocabularies are never invented; they fall back to a general judgment.

Routing interacts with a property of the specialists that we treat as load-bearing: **a specialist's output is not by itself evidence that the queried thing exists**. Some specialist interfaces return their best candidate even when the query has no referent in the clip—a returned time window or bounding box can therefore reflect the query rather than the content. Existence and occurrence decisions are consequently assigned to the general verifier, which sees the frames densely, while the specialists supply the quantities, windows, masks, depths, and tracks it cannot produce. This assignment avoids treating a forced specialist output as proof of occurrence; it does not make absence judgments infallible. Composite claims combine the two: an ordering claim is first confirmed as two occurrences by the verifier, and only then ordered using the specialist's windows.

4.7.5 Executing equivalent measurements once

Sharing happens at two levels, and we keep them distinct. Within a plan, repeated references may share one internal description, so a prompt mentioning the same object in five claims does not describe it five times. At execution, two measurements collapse into a single tool call only when the complete measurement request is identical—the operation, the referent, the exact query, the output kind, and the time range—so a whole-clip measurement and a sub-interval measurement are never merged. This is exact-request reuse, equivalent to common-subexpression elimination in the database sense [41, 42]; no sharing across merely similar queries is performed. Whether this yields a net saving in practice is a question for the evaluation, not a claim made here. Temporal relations remain directional and tolerance-scaled, following a subset of Allen's interval algebra [1].

4.7.6 What the planner does not change

The planner is deliberately the only part that changes: a deterministic translator translates its lines into exactly the plan representation the compiler already consumes, so the compiler, the sharing rules, the measurement contracts, the execution layer, and the division between measuring and judging are untouched. This separation is intended to keep the planner replaceable and to prevent a change of planning style from silently altering how evidence is interpreted.

4.7.7 Validation status and known gaps

The planner has been checked *at the planning stage only*: real model calls, no video, no tools. We define a claim as *measurement-carrying* when its validated plan contains at least one non-generalist operation that survives validation without being salvaged or falling back; a claim combining a general judgment with a count counts once, and salvaged terms never count.

On 233 claims drawn from 22 human-annotated clips, every claim produced a valid, type-correct, compilable plan—no fallbacks, no schema errors, no compile errors—at one to two model calls per clip. Measurement-carrying claims were 27/233 (11.6%), against 2.3% for the earlier tabular format on the same claims: a roughly fivefold relative increase that nonetheless leaves a large majority of these claims resting on a general judgment, because these prompts are dominated by occurrence-style assertions. On two separate diagnostic sets chosen to be measurement-dense, the rate was 19/30 (63.3%) on claims drawn from counting, spatial, and ordering flaws, and 10/10 on synthetic counting prompts. Those diagnostic sets probe the routing rules; they are not an estimate of coverage or verdict accuracy on the evaluation benchmark.

We are explicit about what this does *not* show. No video was processed, so it establishes that plans are valid, typed, compilable, and denser in measurements —*not* that any verdict improves. Effects on flaw coverage, on false accusations, and on cost are *unmeasured*; they are decided by the controlled end-to-end comparison of section 6, which contrasts plan-based execution with independent per-claim verification under matched cost accounting. A comparison between planners alone would not answer that question.

Known gaps: the spatial vocabulary has no "on top of" relation, so such claims degrade to a general judgment; the plan-based path has no dedicated sound-event check, so audio claims are not audio measurements on that path; negated measurements are unsupported by design; and outside the reported set, testing has exposed occasional out-of-vocabulary calls, which the validator converts into charged fallbacks rather than silently accepting.

Where specialists actually run. We distinguish two configurations. In the *routed* configuration each claim is dispatched to its dedicated specialist (table 3) and those tools run as executable stages. In the *plan-based* configuration the compiler and its typed checks run against a tool server that exposes the same specialists; in the runs completed so far, that path resolved most checks through the dense-frame generalist, both because occurrence-style claims route there by design and because the earlier planner rarely requested measurements—the specific limitation the planner described here is intended to address. Global plan selection among alternative tool recipes, and a value-of-information loop that spawns follow-up measurements, remain PROPOSED.

5 The Intended Refinement Loop

The three components are meant to close a loop: the benchmark trains and evaluates the critic; the critic scores generated clips; and the critic's verdict becomes a reward that fine-tunes or guides the generator toward physical faithfulness. **This loop is not yet built.** The generator and the critic today run on disjoint data—the generator on simulation-driven clips, the critic on the human-annotated benchmark—and no reward-driven generation run exists. Prerequisites are a critic whose reward is trustworthy on the frozen human labels (section 6) and a decision about where the reward enters generation (preference optimization, guidance, or data selection). We state the loop as the organizing goal and as explicit future work, not as a result.

6 Evaluation Protocol and Measurement

This section specifies how the critic will be measured; it deliberately reports no accuracy or coverage numbers.

Target and ground truth. The frozen physics-focused set of section 3 (150 clips / 306 flaws for routine measurement, 606 clips / 1,107 flaws for full sweeps). Human labels are the only ground truth. The 150-clip / 306-flaw core is the optimization set: because design decisions are repeatedly made against it, reported coverage measures fit to that core rather than generalization, and a generalization claim requires a separately sequestered held-out set that has not influenced system design.

Matching is outside the critic. To decide whether a critic output corresponds to a human-flagged flaw, a separate protocol uses two judge models (a GPT-family and a Gemini-family model, majority over runs). These judges perform *flaw matching only*; they are not components of the critic, do not produce its verdict, and are not treated as ground truth. Any "majority-of-three" in this project refers to such a model-vote protocol, never to human annotators.

Recall, and the false-alarm gap. Because the benchmark has no clean clips, the primary measurable quantity is recall of the human flaws. A meaningful false-alarm or precision number requires either clean clips or per-allegation adjudication of the critic's non-matching outputs; we specify that adjudication as part of the protocol and do not substitute a recall number for it.

Dense-frame requirement. The critic must observe the video densely. The evaluated components extract frames densely under a logged sampling policy, and any measurement obtained from a sparsely sampled input is excluded. A re-run counts only if it uses the dense path on the unchanged frozen target with BF16 tools at pinned revisions.

The frozen confirmatory comparison. We evaluate the orchestration design on the 150-clip optimization set, which carries all 306 human-labeled flaws. The primary paired comparison estimates the joint effect of structured execution—shared measurements, routing to capabilities, composition across measurements, and conflict resolution—relative to a wording-controlled per-claim baseline on the same complete set. Because verification questions can be phrased in ways that produce different verifier behavior, the per-claim baseline uses fixed wording. An auxiliary control runs the per-claim baseline's exact claim texts through the plan machinery; in the current build those claim texts are byte-identical, so this control measures added plan cost and run-to-run serving variability and cannot test a decomposition effect. A separate test of decomposition would require distinct claim texts and is deferred.

Because a detector call, a tracker call, and a full vision-language judgment have different costs, we record model and tool calls, processed frames, and GPU-seconds rather than counting every invocation as equally costly. We report each system at its uncapped operating cost and make cost-matched comparisons in both directions: the structured system is capped at the per-claim baseline's cost, and the per-claim baseline is extended to the structured system's uncapped cost, using a priority policy fixed in advance. Allegations that do not match a human-labeled flaw count as *review burden*, not false positives, in the frozen-benchmark score; any separate adjudication of those allegations is reported only as an analysis of possible omissions in the benchmark labels, and does not expand the frozen labels or alter the frozen-benchmark score. We estimate uncertainty by resampling whole clips, because flaws within the same clip are dependent.

No comparative results are reported here. The final report will disclose that this set was inspected and used during optimization, that the wording-controlled per-claim baseline was introduced after treatment outputs had been observed, and that no retroactive held-out split exists within these 150 clips; a claim of generalization therefore requires evaluation on a new frozen set.

7 Related Work

7.1 Physics and structure in video generation

Physical-commonsense generation and its evaluation are studied by VideoPhy [5, 6], PhyGenBench [36], and Physics-IQ [38]; V-JEPA-style work argues physics is best judged in a learned state space [14]. Our generator does not learn physics: it *imports* it from a simulator and asks the backbone to render it, via structure conditioning. Structure- and motion-conditioned synthesis has a strong lineage—ControlNet for images [57], motion control for video [49], the Wan family and its VACE control interface [25, 48], and, as guided editing rather than control-branch conditioning, SDEdit [35]. DiffPhy conditions a diffusion backbone on a physics chain-of-thought [56]; we instead condition on a rendered depth field from a simulation.

7.2 Evaluating text-to-visual faithfulness

Decomposing a prompt into checkable pieces is established: TIFA [22] and the Davidsonian Scene Graph [9] score generated questions/propositions with a vision-language model, VQAScore uses an image-to-text model [31], and GenAI-Bench, VBench, T2V-CompBench, FETV, and EvalCrafter benchmark compositional alignment [23, 28, 32, 33, 45]; embedding and preference scorers predict alignment or human preference [21, 27, 53], and learned quality models predict human judgment directly [19, 20]. We keep such a holistic learned scorer out of the verdict path. Relative to these, our critic routes each claim to a dedicated open perception specialist under an evidence-only contract, validates a typed plan before execution, distinguishes event occurrences, reuses perception only under an exact contract, and preserves claim-level provenance with an explicit unknown outcome. We claim novelty for the integration, not the parts.

7.3 Visual programs, tool-use, and modular video QA

Composing perception modules for reasoning includes Neural Module Networks [4], VisProg [17], ViperGPT [46], CodeVQA [44], and neural-symbolic VQA [24, 55]; LLM tool-controllers and plan-execute graphs include HuggingGPT, TaskMatrix.AI, ControlLLM, LLMCompiler, and ReWOO [26, 30, 34, 43, 52], with search-based variants [7, 54, 60, 61]; modular video QA includes ProViQ, MoReVQA, the VideoAgent systems, and VideoTree [10, 12, 37, 50, 51]. Our execution layer is narrower and verification-specific: a *typed* plan, type-checked before tools run, whose evidence obligations carry three-valued semantics; it does not search over alternative plans. Reuse of equivalent measurements is common-subexpression elimination in the multi-query-optimization sense [15, 16, 41, 42], and the specialists themselves draw on open detection, counting, tracking, grounding, and audio models [2, 3, 8, 11, 13, 18, 29, 40, 58, 59].

7.4 Flaw and preference datasets

Prior video benchmarks largely provide clip-level physical-commonsense or preference labels [5, 6, 36]. Our benchmark instead provides *localized* human flaw annotations—severity, time span, per-frame boxes, the violated prompt fragment, and a rationale—so that a detector can be scored not only on whether a clip is flawed but on whether it found the specific, spatially and temporally located error a person marked.

8 Limitations

Generation. Validation is qualitative (visual inspection plus a trajectory tracker); there is no learned quality metric for this line. Multi-object interaction is unsolved. The control signal comes from a simulator, so scene diversity is bounded by what is simulated, and the mapping from simulation to prompt appearance is not itself verified for physical consistency.

Benchmark. The corpus is recall-only (no clean clips), so precision is not directly measurable; flaws are single-annotator with no inter-annotator agreement; the scope/category tags are machine-derived; and the source generator of the clips is unrecorded. Repeatedly designing against one frozen set risks overfitting to it even without training on it.

Critic. The spatial relation predicate and negative-event verification are limited; the typed-plan layer does not yet dispatch the full specialist registry; binding remains a central failure mode, since the compiler enforces symbolic identifier consistency but cannot guarantee that heterogeneous tools localized the same physical instance; and the reasoner-mediated sub-checks reduce but do not remove semantic opacity.

Loop. The generator–critic loop is unbuilt, and critic quality on the frozen labels is the gate before any reward-driven generation is attempted.

9 Conclusion

Physical faithfulness in generated video requires steering, measurement, and ground truth together. This report contributes one piece for each: a generator that imports geometry and motion from a simulation and renders them through a video-diffusion control branch; a human-annotated benchmark that localizes real flaws in space, time, and prompt reference; and an auditable multi-tool critic that decomposes a prompt, routes each claim to an open specialist returning evidence only, and combines the evidence deterministically into a three-valued verdict. We have described each component and its honest implementation status and specified the human-label protocol under which the critic—and eventually the loop that lets it refine the generator—will be measured. The comparative numbers, and the closed loop itself, are the next milestones, not claims of this report.

A Appendix

To be expanded: the full generation configuration and control-render details; the benchmark schema and per-stratum statistics; the critic's plan schema, predicate type rules, and temporal-relation truth tables; the specialist license and revision manifest; and the pre-registered evaluation manifest.

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